

WASP-12b

R-0405

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230116 · created 2026-07-19T05:05:04+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459961.7065 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.03
Transit depth (Rp/Rs)^2	2.39 +/- 0.92 [%]
Orbital Inclination (inc)	77.07 +/- 2.83
Airmass coefficient 1 (a1)	1.052 +/- 0.037
Airmass coefficient 2 (a2)	-0.042 +/- 0.033
Scatter in the residuals of the lightcurve fit is	3.49 %
Best Comparison Star	#2 - [418, 141]
Optimal Method	PSF photometry
Transit Duration (day)	0.109 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.03 vs 0.1178 (deviation 1.24 σ)
Duration fit vs published	0.109 d vs 0.125 d (deviation 1.14 σ)
Mid-time O–C	1568.8 min
Depth SNR	5.2
Residual scatter	3.49 %

Light curve

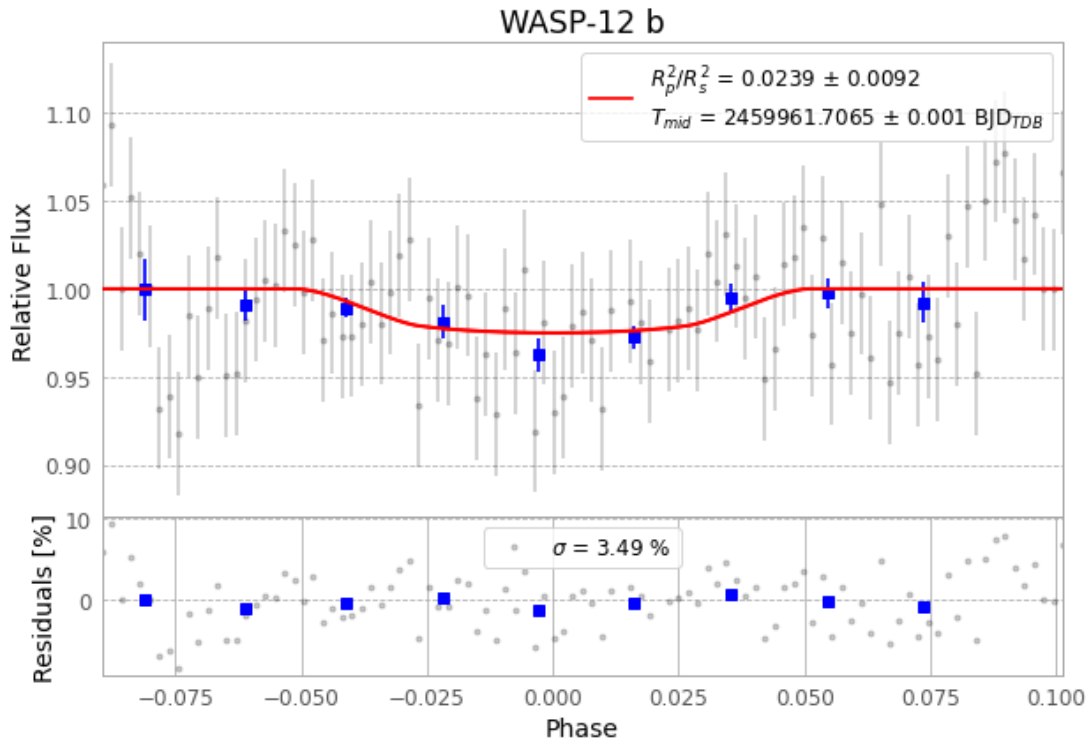


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-16.png (SHA-256 b36ea937969c8c77...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [159, 13], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 196d37244bfb7032...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

101 FITS frames (66 MB), WASP-12230117022715.FITS → WASP-12230117072716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230116.log (38fa4cb74ed3...), FinalLightCurve_WASP-12 b_2023-01-16.png (b36ea937969c...), FinalLightCurve_WASP-12 b_2023-01-16.csv (25e52435d0bc...)

Compute resources

Wall time 891.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 05:05Z.